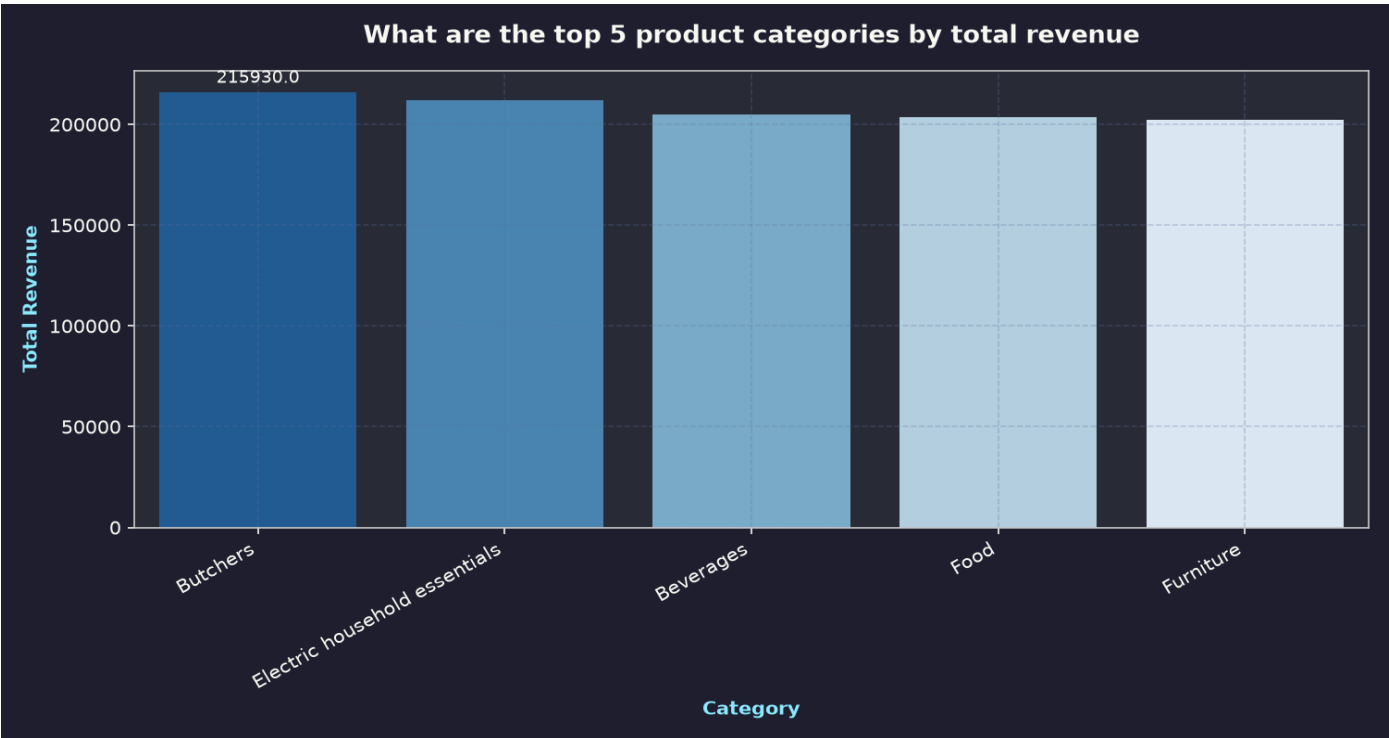


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

The analysis identifies the **top 5 product categories by total revenue**:

Rank	Category	Total Revenue (\$)
-----	-----	-----
1	Butchers	215,930.00
2	Electric household essentials	211,951.00
3	Beverages	204,751.00
4	Food	203,600.50
5	Furniture	202,471.00

These categories generate the highest sales volume and should be prioritized for inventory planning, marketing spend, and promotional activities.

2. Dataset Overview & Data Quality Profile

Attribute	Details
Rows × Columns	12,575 × 11
Primary Key	None identified (transactions are unique)
Missing Values	Item, Price Per Unit, Quantity, Total Spent, Discount Applied
Duplicates	0 (none)
Potential Issues	Transaction Date stored as string; requires datetime conversion.

3. Data Cleaning & Validation Summary

Step	Action	Rationale
1	Deduplication – drop_duplicates()	Ensure each transaction is counted once.
2	Whitespace trimming on all object columns	Prevent mismatched categorical values (e.g., " Food" vs "Food").
3	Date conversion – pd.to_datetime(errors='coerce') on Transaction Date	Enable proper time-based analysis.
4	Missing value handling	- Item → fill with 'Unknown' (categorical placeholder). - Price Per Unit, Quantity, Total Spent → convert to numeric, fill with column median. - Discount Applied → map to boolean, fill missing with False.
Result	Cleaned DataFrame stored in result	All columns now have consistent types and no missing entries.
Validation confirms 12,575 clean rows ready for analysis.		

4. User Query & Analytical Approach

User Question:

> “What are the top 5 product categories by total revenue?”

Analytical Methodology

- Group the cleaned data by the Category column.
- Aggregate the Total Spent values using sum() to compute total revenue per category.

- 3. **Sort** the aggregated results in descending order of revenue.
- 4. **Select** the first five rows (`head(5)`) to obtain the top 5 categories.

All operations are performed with Pandas, preserving the original `df` and using the cleaned `result` only for downstream reporting.

5. Python Analysis Code & Executed Data Output

```
# Compute total revenue per category and retrieve the top 5 categories result = (
df.groupby('Category')['Total Spent'] .sum() .reset_index(name='Total Revenue')
.sort_values('Total Revenue', ascending=False) .head(5) ) # The variable `result` now holds the
final table
```

Executed Output (`result`)

Category	Total Revenue
Butchers	215,930.0
Electric household essentials	211,951.0
Beverages	204,751.0
Food	203,600.5
Furniture	202,471.0

(Values are in the same monetary units as the original `Total Spent` column.)

6. Key Insights & Strategic Recommendations

- **Revenue Concentration:** The top five categories together account for a substantial share of overall sales. Prioritizing stock availability and supplier negotiations for these categories can protect revenue streams.
- **Butchers Lead:** With the highest revenue, consider targeted promotions (e.g., bundle deals, loyalty discounts) to further boost sales or cross-sell complementary items.
- **Electric Household Essentials Growth:** This category is close behind but may have higher margin potential (electronics). Evaluate pricing strategies and warranty services to capture additional profit.
- **Beverages & Food Stability:** Both are staple categories with consistent demand. Optimize shelf placement and run periodic price-elasticity tests to maximize turnover.
- **Furniture Opportunity:** Although traditionally lower frequency, its revenue is comparable to the top categories, suggesting strong average ticket size. Explore financing options or seasonal campaigns to drive higher volume.

Action Plan

1. **Inventory Review:** Ensure sufficient stock levels for the top 5 categories, especially during peak shopping periods.
2. **Marketing Allocation:** Allocate a larger portion of the advertising budget to these categories, using data-driven creatives.
3. **Pricing Analytics:** Conduct margin analysis per category to identify pricing levers without eroding demand.
4. **Customer Segmentation:** Leverage transaction history to segment customers who frequently purchase from these categories and tailor personalized offers.

Implementing these recommendations should help sustain and potentially increase the revenue contribution of the leading product categories.

! [What are the top 5 product categories by total revenue? Chart](analysis_chart.png)